

# An Ensemble Detector

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Chapter 4 of the thesis

## From One Detector to Many

Chapter 3 ended on a diagnosis: ReReReRe is a strong detector of *inconsistent* careless responding but is, by design, nearly blind to *consistent* patterns such as straightlining and acquiescence, weak at low dimensionality, and—like any single index—limited by the need to commit to one threshold. Each of these limits is the mirror image of a strength of some other detector. Longstring and low intra-individual variability catch straightliners outright; Mahalanobis distance catches multivariate outliers. The detectors are *complementary*: they fail on different respondents.

This chapter turns that complementarity into a single, better detector. Rather than choosing one index or applying several under separate ad hoc cutoffs—the informal multi-index practice that Chapter 1 described as the field’s de facto recommendation (Curran, 2016; Ward & Meade, 2023)—it treats each index as one *feature* and lets a trained classifier learn how to weigh them and when to trust each. The classifier is a Random Forest (Breiman, 2001), and the central empirical question of the chapter is whether ReReReRe contributes anything *beyond* the established auxiliary detectors, and if so, how much and under what conditions. The answer, developed with bootstrap confidence intervals, is that its contribution is statistically positive at every questionnaire length and grows substantially with length.

## The Five Features

Each respondent is summarized by five detector scores, all oriented so that *higher means more careless*:

**$z_{RR}$**  the ReReReRe permutation coherence z-score of Chapter 3, sign-flipped so that low coherence (incoherent, inconsistent careless) reads as a high feature value.

**IRV** the (negated) intra-individual response variability (Dunn et al., 2018): low dispersion—the signature of straightlining—reads as high careless.

**LongString** the longest run of identical consecutive responses (Johnson, 2005).

**$D^2$**  the Mahalanobis distance from the sample centroid (Mahalanobis, 1936): multivariate outlyingness.

**Person–Total** the (negated) correlation between a respondent’s profile and the sample mean profile: answering against the grain.

The first feature is the contribution of this thesis; the remaining four are the established indices reviewed in Chapter 1. The design intention is explicit division of labour:

ReReReRe carries the *inconsistent*-careless signal that the four auxiliaries miss, while the auxiliaries carry the *consistent* signal that ReReReRe misses. Whether that intention is realized is tested in Section 4.6.

### The Random Forest Combiner

The five features are combined by a Random Forest (Section 2.8): an ensemble of 200 decision trees, each grown on a bootstrap resample of the training respondents, each split considering a random subset of the features, the final predicted careless probability being the proportion of trees voting “careless.” Two properties make the Random Forest the right combiner here. First, the features interact: a respondent with both a low  $z_{RR}$  and a low IRV is far more suspicious than the additive combination of the two signals implies, because that conjunction is the signature of a straightliner who also breaks cross-factor coherence. Trees capture such interactions automatically. Second, the Random Forest is robust to features on different scales and to redundancy among them, so the five detectors can be fed in raw. In our simulations the Random Forest outperformed logistic regression and elastic-net regression on every evaluation metric considered, with the widest margins on the hardest conditions (short questionnaires and low base rates), consistent with the non-linear, interactive structure of the problem.

Throughout, the classifier is evaluated by five-fold cross-validation (Section 2.9): each respondent receives an out-of-fold predicted probability from a forest that did not see them in training, and these out-of-fold predictions are used for every metric reported below. This guards against the optimism that would arise from scoring the classifier on its own training data.

### Operational Definition of Careless

Evaluating a detector requires a ground-truth label, and in simulated data a respondent’s corruption can be set to any degree. We therefore need an explicit rule for *how corrupted* a respondent must be to count as careless. This rule is not innocuous: set it too high and the negative class is contaminated with heavily corrupted respondents; set it too low and trivially-corrupted respondents inflate the positive class.

We fix the rule empirically. A respondent is labelled careless when the corrupted fraction of their row exceeds  $\tau = 0.40$ , and *every* other respondent—clean and lightly corrupted alike—stays in the negative class. The cutoff  $\tau = 0.40$  is the joint argmax of the Matthews correlation coefficient, Cohen’s  $\kappa$ , Youden’s J, and balanced accuracy across a sweep  $\tau \in \{0.10, 0.20, \dots, 0.90\}$  evaluated on 24,000 simulated respondents. (Two secondary metrics, F1 and AUPRC, favour slightly lower cutoffs of 0.30 and 0.20; the four primary classification metrics agree unanimously on 0.40.) Crucially, this is the *deployment-realistic* framing: in real data the analyst does not know in advance who has 30% versus 60% corruption, so keeping the lightly-corrupted respondents in the negative class tests whether the detector recovers moderately-to-fully corrupted rows without false-flagging the ambiguous middle. An earlier, looser definition that excluded the middle from the negative class was discarded as artificially easy.

**Table 1**

*Pooled detection performance of the five-feature ensemble under the rate-aware operating point (160 cells, five-fold CV within each).*

| Metric      | Pooled value |
|-------------|--------------|
| MCC         | 0.662        |
| F1          | 0.730        |
| Sensitivity | 0.731        |
| Specificity | 0.932        |
| AUC         | 0.919        |

**Table 2**

*Per-size detection (MCC) under the rate-aware operating point.*

| Items      | 30    | 50    | 80    | 100   | 150   | 200   | 250   | 300   |
|------------|-------|-------|-------|-------|-------|-------|-------|-------|
| <b>MCC</b> | 0.439 | 0.525 | 0.604 | 0.630 | 0.717 | 0.771 | 0.798 | 0.815 |

### Simulation Design and Headline Performance

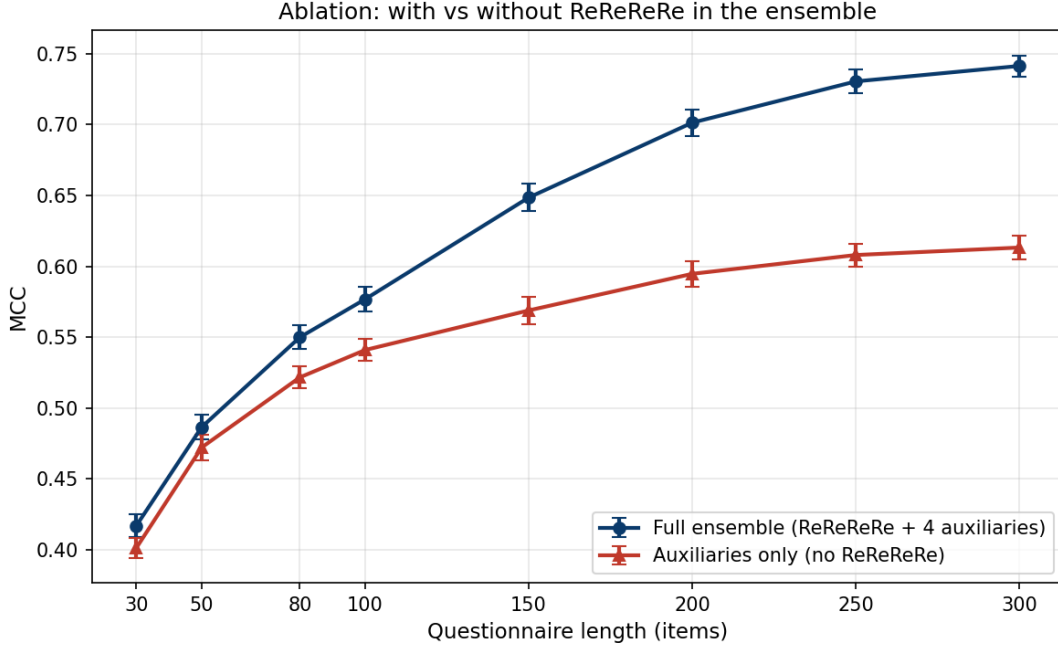
The ensemble was evaluated on a simulation grid of **384 datasets** crossing eight questionnaire sizes (30, 50, 80, 100, 150, 200, 250, and 300 items), four careless base rates (10%, 20%, 40%, 60%), and twelve replications per cell, with  $n = 500$  respondents each. Within every dataset, careless respondents were injected at one of six prototypical patterns—random, longstring, mixed, fatigue, acquiescent, pure straightlining—with the corrupted fraction drawn from  $\{0\%, 10\%, \dots, 100\%\}$ , so that the full corruption continuum is represented. The five features were computed for every respondent and a Random Forest trained under five-fold cross-validation.

The operating point is set by the *rate-aware* calibration, the deployable rule developed in Chapter 5: the analyst supplies a rough prior on the careless base rate and the threshold is placed at the corresponding quantile of the predicted probabilities, requiring no labels. Pooled across all sizes and rates, the ensemble reaches the performance in Table 1.

Detection quality scales smoothly with questionnaire length (Table 2). At 30 items the ensemble is useful but modest ( $\text{MCC} = 0.44$ ); by 150 items it is strong ( $\text{MCC} = 0.72$ ); and it approaches its asymptote near 300 items ( $\text{MCC} = 0.82$ ), where the  $250 \rightarrow 300$  gain is only  $+0.017$ . This is the same total-length scaling identified for ReReReRe alone in Chapter 3, now inherited by the ensemble.

### The Contribution of ReReReRe

The chapter’s central question is whether ReReReRe earns its place in the ensemble. We answer it by *ablation*: training the same Random Forest under two feature sets—the full five features, and the four auxiliaries only—and comparing them under the identical cross-validation protocol and operational definition.

**Figure 1**

*Ablation by questionnaire length. The full five-feature ensemble (ReReReRe + four auxiliaries) versus the four auxiliaries alone, both trained under five-fold cross-validation with the same operational definition. Error bars are  $\pm 1$  SEM. The gap grows from a small margin at 30 items to a sizeable one at 200–300 items.*

### Ablation by questionnaire length

Figure 1 shows the two configurations as a function of questionnaire length. The full ensemble (with ReReReRe) rises from  $\text{MCC} \approx 0.42$  at 30 items to  $\approx 0.74$  at 300; the auxiliaries-only ensemble rises from  $\approx 0.40$  to only  $\approx 0.61$  and then plateaus. The gap between the curves *widens monotonically* with length: at short questionnaires the four auxiliaries already capture most of the available signal, but at long questionnaires they saturate while the full ensemble keeps improving. Adding more auxiliary detectors does not close this gap—it is specifically the cross-factor coherence signal, available only from ReReReRe, that the auxiliaries cannot supply.

### Bootstrap confidence intervals on the contribution

A difference of curves is suggestive; to make the claim inferential we put a confidence interval on it. For each questionnaire size we bootstrapped over respondent indices (500 resamples; Section 2.7) and recomputed  $\Delta\text{MCC} = \text{MCC}(\text{full}) - \text{MCC}(\text{auxiliaries only})$  on each resample, taking the 2.5th and 97.5th percentiles as a 95% interval. Table 3 and Figure 2 report the result.

The interval *excludes zero at every size tested*, so the contribution of ReReReRe is statistically positive throughout—including the 30-item case, where one might have ex-

**Table 3**

*Bootstrap 95% confidence intervals on the contribution of ReReReRe,  $\Delta MCC = MCC(full) - MCC(auxiliaries\ only)$ , by questionnaire length.*

| Items | $\Delta MCC$ | 95% CI           | $P(\Delta > 0)$ |
|-------|--------------|------------------|-----------------|
| 30    | +0.020       | [+0.009, +0.031] | 1.000           |
| 50    | +0.018       | [+0.008, +0.028] | 1.000           |
| 80    | +0.035       | [+0.024, +0.046] | 1.000           |
| 100   | +0.035       | [+0.025, +0.045] | 1.000           |
| 150   | +0.096       | [+0.085, +0.110] | 1.000           |
| 200   | +0.121       | [+0.108, +0.133] | 1.000           |
| 250   | +0.133       | [+0.120, +0.145] | 1.000           |
| 300   | +0.150       | [+0.138, +0.163] | 1.000           |

pected it to be redundant. The *magnitude*, however, is strongly length-dependent: small below 100 items ( $\approx +0.02$  to  $+0.035$ ) and large above 150 items ( $+0.10$  to  $+0.15$ , roughly a quarter of the auxiliaries’ ceiling). This is the chapter’s headline structural finding, and it has a clear practical reading: ReReReRe pays off most on the long multi-construct batteries that dominate modern psychometric practice (150–300 items), not on the short unidimensional scales where it would be easiest to dismiss as redundant.

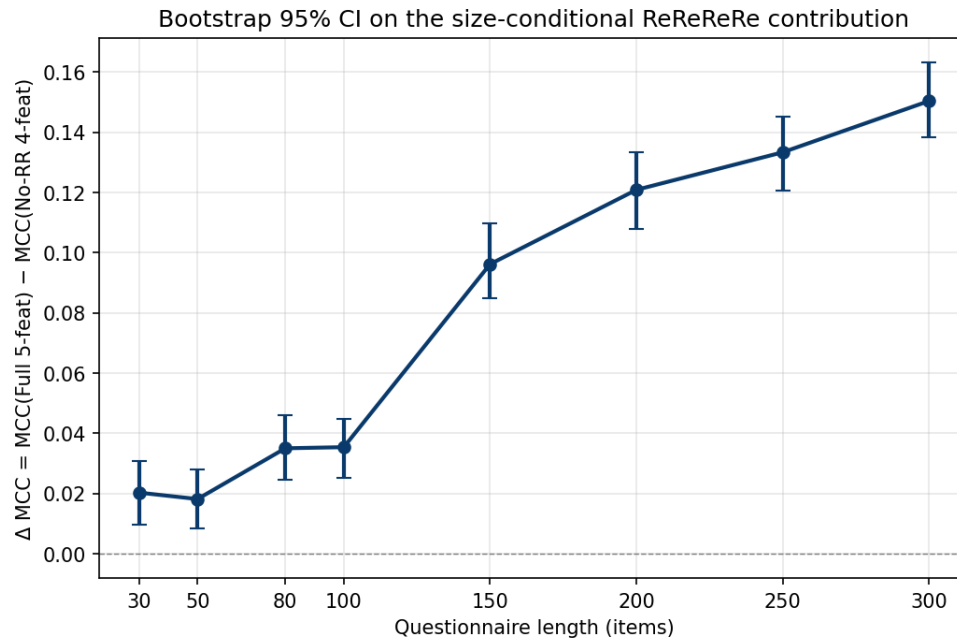
### Where the Contribution Comes From

The ablation says *how much* ReReReRe contributes; the per-pattern breakdown says *where*. Figure 3 plots detection rate against corruption fraction for each of the six injected patterns, comparing the full ensemble with the auxiliaries-only ensemble.

The picture is exactly the complementarity the ensemble was designed to exploit. On the *consistent* patterns—pure straightlining, acquiescence, and highly-corrupted longstring and fatigue—the two curves are essentially identical: the auxiliary detectors already saturate, and ReReReRe adds nothing, because a respondent who selects the same option throughout is trivially caught by LongString and low IRV. On the *inconsistent* patterns—mixed and, especially, random—the full ensemble (green) sits well above the auxiliaries-only ensemble (red) across the whole corruption range: here the auxiliaries are partly blind, and ReReReRe rescues respondents they miss. The ensemble works not because every detector helps everywhere, but because the detectors fail on disjoint sets of respondents, and the Random Forest learns to route each case to whichever feature is informative for it.

### Summary

Combining ReReReRe with four established auxiliary detectors inside a Random Forest yields a single careless-detection model whose pooled performance ( $MCC = 0.66$ ,  $AUC = 0.92$ ) scales from  $MCC = 0.44$  on 30-item questionnaires to  $MCC = 0.82$  on 300-item batteries under a deployable, label-free operating point. Ablation with bootstrap confidence intervals establishes that the contribution of ReReReRe is statistically positive at every questionnaire length and grows from  $+0.02$  MCC on short instruments to  $+0.15$  MCC on long ones. The mechanism is complementarity: ReReReRe rescues the inconsistent careless

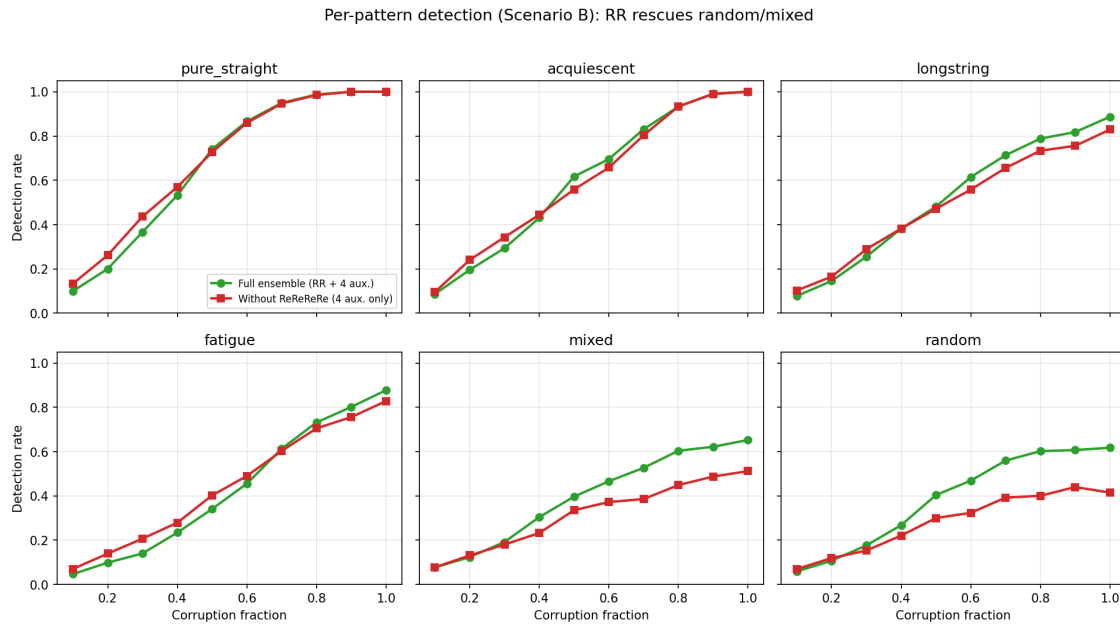
**Figure 2**

*Bootstrap 95% confidence interval on the size-conditional contribution of ReReReRe ( $\Delta \text{MCC}$  = full – auxiliaries only). The interval excludes zero at every questionnaire length; the contribution grows from +0.020 at 30 items to +0.150 at 300 items.*

patterns (random, mixed) on which the auxiliary detectors are weak, while contributing little on the consistent patterns they already catch. What remains is the question the headline numbers quietly presupposed—how to set the operating point *without* the ground-truth labels that the cross-validation here had access to. That is the deployment problem, and it is the subject of Chapter 5.

## References

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**Figure 3**

*Per-pattern detection: full ensemble (green) versus auxiliaries only (red), by corruption fraction. The two coincide on consistent patterns (pure straightlining, acquiescence, longstring, fatigue), which the auxiliaries already detect; they diverge on the inconsistent patterns (mixed, random), where ReReReRe supplies the signal the auxiliaries lack.*

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